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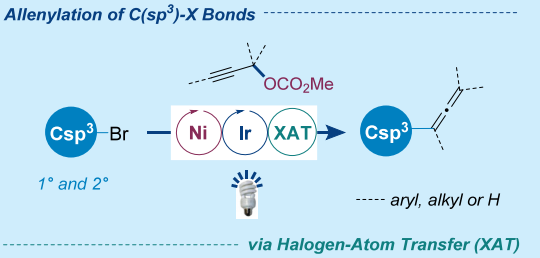
# Nickel-Catalyzed Photoredox Allenylation of Alkyl Halides

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**ABSTRACT:** We report a dual Ni/photoredox-catalyzed cross-coupling method for propargyl carbonates and nonactivated alkyl bromides, facilitating the synthesis of a variety of substituted allenes under mild and practical conditions. Mechanistically, the reaction integrates Ni-catalyzed activation of the propargyl electrophile via $S_N2'$ oxidative addition at Ni(I) with silyl radical-induced activation of the alkyl halide through halogen-atom transfer. This methodology provides a gentle approach for introducing allenyl groups into complex halogenated aliphatic molecules, offering further opportunities for derivatization.




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Allenes represent a highly versatile class of compounds exhibiting unique physical and chemical properties.[1] These characteristics make allenes valuable building blocks in organic synthesis,[2] contributing to the construction of larger and structurally complex molecular targets, including natural products, pharmaceuticals,[3] and advanced materials.[4] Therefore, developing robust and mild strategies for the catalytic allenylation of organic molecules remains a significant and active area of research in synthetic chemistry.[5] Early developments in transition metal-catalyzed allenylation methods primarily focused on palladium-catalyzed $S_N2'$-type nucleophilic substitution reactions of propargyl electrophiles.[6] However, these methods often face challenges such as poor functional group tolerance when using highly reactive organometallic nucleophiles, and regioselectivity issues on possible formation of the isomeric propargylation products. Despite substantial progress in developing efficient strategies for the allenylation of unsaturated substrates, particularly those involving C($sp^2$)-X bonds, the involvement of aliphatic coupling partners in these processes remains challenging.

Alkyl halides are abundant and easily accessible precursors of organometallic aliphatic carbons. The nucleophilic addition of zinc or magnesium organometallics to propargyl electrophiles has been well investigated under various transition metal catalysts, including palladium, copper, iron,[7] and more recently, nickel[8] (Scheme 1a). However, developing new methods for the allenylation of nonactivated alkyl halides through direct C($sp^3$)-X bond manipulation would be exceptionally advantageous. Such methods would avoid preparation and handling of air- and moisture-sensitive organometallic reagents, allowing better functional group tolerance and enabling the use of more complex organohalides.

To date, there is only one prior report by Wang describing a nickel-catalyzed cross-electrophile coupling of propargyl electrophiles with nonactivated alkyl halides.[9] While this protocol allows for the construction of a wide range of allenic frameworks, it notably excludes all-aliphatic allenes and only applies to alkyl iodide substrates, which are less attractive due to their limited availability and stability compared to their bromo- and chloro-counterparts. Additionally, this method requires the use of excess of manganese metal as a reductant and proceeds best in *N*-methyl-2-pyrrolidone, a less desirable solvent in terms of sustainability and toxicity (Scheme 1b).

Herein, we report a nickel/photoredox dual catalytic strategy for the selective allenylation of unactivated alkyl bromides based on silyl radical-mediated halogen-atom transfer (XAT) process (Scheme 1c).[10] In this approach, a silane functions as both a XAT promoter and a reductive quencher of the photocatalytic cycle. This method provides facile access to diversely functionalized tri- and tetrasubstituted allenic compounds,[11] including all-aliphatic derivatives, under very mild conditions without the need for air-free techniques. Moreover, the reaction can be conducted in acetone, a more sustainable and less toxic solvent. The strategy also avoids the need for stoichiometric metal-based reagents or reductants. A mechanistic rationale for the cross-electrophile coupling reaction is provided.

Preliminary studies focused on the cross-coupling of 4-bromo-tetrahydro-2*H*-pyran (**1a**) with alk-2-ynyl carbonate (**2a**) in the presence of a stoichiometric amount of $(TMS)_3SiH$,[12] as a model reaction (Table 1). First experiments led to the identification of nickel(II)$Cl_2$.glyme

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**Scheme 1. Strategies for the Allenylation of Alkyl Halides with Propargyl Electrophiles under Nickel Catalysis**

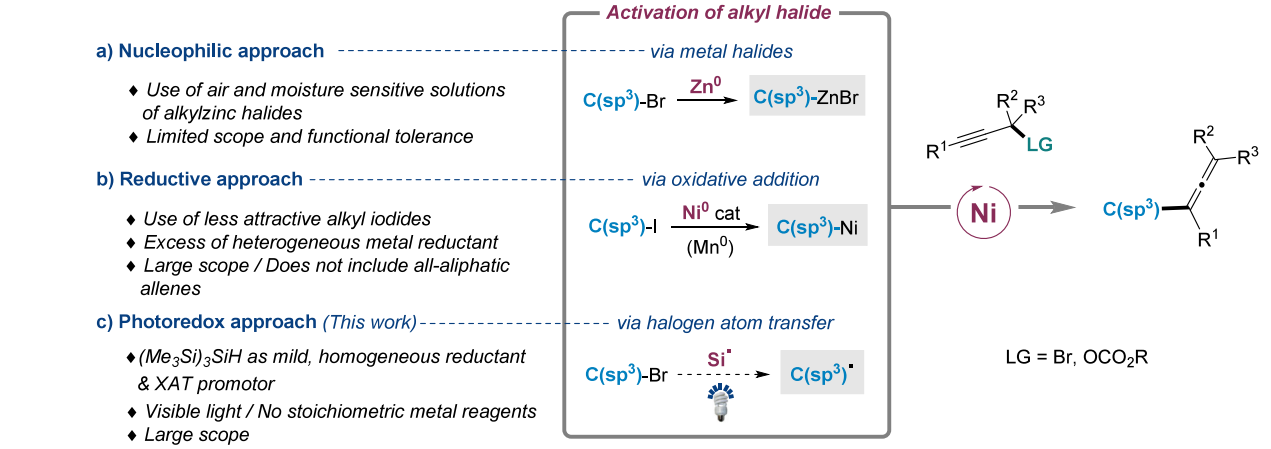
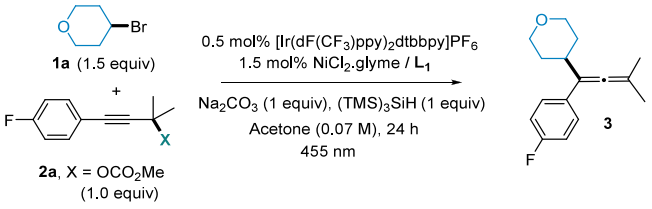
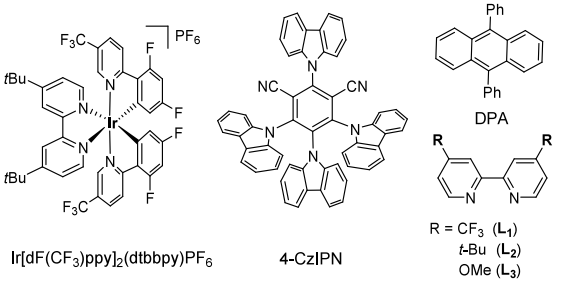



**Table 1. Selected Optimization Experiments**[a]




| Entry | Deviation from above reaction | Yield (%)[b] |
|---|---|---|
| 1 | None | 84 (0) |
| 2 | No nickel/Ligand | 0 |
| 3 | No ligand | 15 (79) |
| 4 | No photocatalyst (LED light only) | 0 |
| 5 | No light (dark) | 0 |
| 6 | No silane | 0 |
| 7 | No base | 0 |
| 8 | $L_2$ as ligand | 51 (0) |
| 9 | $L_3$ as ligand | 65 (0) |
| 10 | 1,10-Phen as ligand | 40 (33) |
| 11 | 4-CzIPN as PC | 63 (0) |
| 12 | DPA (5 mol%) as PC[c] | 68 (0) |
| 13 | $Et_3N$ as base | 0 (0) |
| 14 | X = OBoc | 81 (0) |
| 15 | X = OAc | 17 (40) |




[a]Reactions were performed on a 0.25 mmol scale using a 40 W blue LED Kessil lamp (455 nm) at 50% light intensity. Ppy = phenylpyridine; dtbbpy = 4,4′-di-*tert*-butyl-2,2′-dipyridyl; 4-CzIPN = 1,2,3,5-tetrakis(carbazol-9-yl)-4,6-dicyanobenzene. DPA = 9,10-diphenylanthracene. [b]Yields of **3** were determined by $^{19}F$ NMR spectroscopy against fluorobenzene. Recovery of **2a** in parentheses. [c]A 18 W purple LED lamp (405 nm) was used.

and 4,4′-bis(trifluoromethyl)-2,2′-bipyridine ($L_1$), as effective nickel source and ligand (1.5 mol %), respectively. The iridium complex Ir[(dF($CF_3$)ppy)$_2$(dtbbpy)]$PF_6$ (0.5 mol %) proved to be a suitable photocatalyst for this transformation under the irradiation of a 40 W blue LED lamp (455 nm) at 50% light intensity.[13] Control experiments confirmed that the dual catalytic system, light irradiation, and silane are essential for this transformation (Table 1, entries 2−6). A base was also required (Table 1, entry 7), with sodium carbonate (1 equiv) being the most effective. Finally, acetone was selected as a suitable and sustainable solvent (0.07 M). Under these conditions, the corresponding allene (**3**) was produced in 84% spectroscopic yield within 24 h. Other bipyridine ligands ($L_2$, $L_3$) and organophotocatalysts (e.g. 4-CzIPN and DPA) showed lower performance at similar catalyst loadings (Table 1, entries 8−12). Amine base, such as triethylamine, failed to promote the reaction. Instead, the corresponding allenic reduction product[14] was obtained selectively (Table 1, entry 13). Finally, we identified the OBoc group as possible alternative leaving group, while the OAc group showed much lower efficiency (Table 1, entries 14−15).

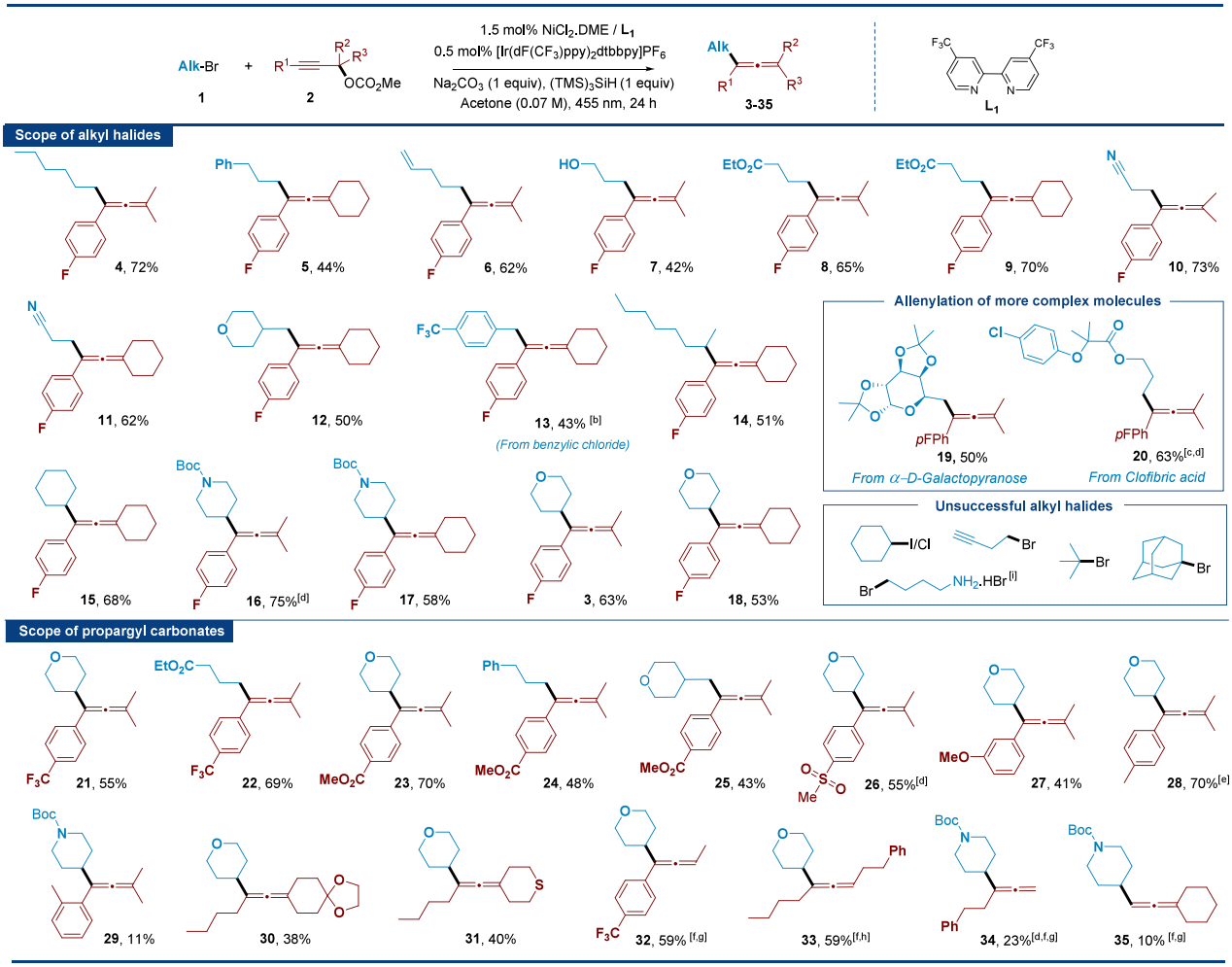
The scope of the reaction was then investigated across a series of propargyl carbonates and readily available alkyl halides (Table 2). The transformation was effective with primary and secondary nonactivated alkyl bromides bearing various functional groups, such as alkene, ether, free alcohol, amine, ester, nitrile, aryl chloride, and fluoroalkyl (**3**−**20**). Notably, alkyl bromides performed better than their chloro- and iodo-counterparts, as demonstrated in the synthesis of allene (**15**) using different cyclohexyl halides. The cyclohexyl allene was obtained in 1%, 71%, and 5% NMR yields, from the iodo-, bromo-, and chloro-alkyl coupling partner, respectively.[15] However, activated benzyl halides were an exception. While benzyl bromides faced challenges in the allenylation process, the more desirable and readily available benzylic chlorides proved better coupling partners, as illustrated in the synthesis of benzylic allene (**13**). Noteworthily, tertiary bromides, e.g. *tert*-butyl bromide and 1-bromoadamantane, did not participate in the coupling reaction.

The allenylation of more complex organohalides was also successful. A galactopyranose-derived bromide was cross-coupled to yield the desired allenyl derivative (**19**) in 50% yield. Additionally, a clofibric acid bromoalkyl ester was allenylated in 63% yield (**20**), withstanding the interchange of



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**Table 2. Scope of the Allenylation Reaction**[a]




[a]Reactions performed on a 0.4 mmol scale. A 40 W blue LED lamp (455 nm) used at 50% light intensity. Isolated yields are given. [b]48 h. An additional amount (1.5 mol %) of Ni/L catalyst was added after 24 h. [c]Equimolar amounts of alkyl bromide and propargyl carbonate. [d]0.2 mmol scale. [e]0.3 mmol scale. [f]3 mol % of $NiCl_2$·dme/dtbbpy ($L_2$) were used as catalyst/ligand system. [g]28 h. [h]40 h. [i]2.5 equiv of base.

the alkoxy chain with methoxide generated in the reaction medium.

The scope of propargyl carbonates was further investigated. Tertiary carbonate derivatives with electron-poor or electron-rich aryl substituents (**21**−**28**) performed equally well. Noteworthily, an *ortho* substituent on the aryl rings had a detrimental effect on the reaction efficiency, delivering the desired allene in low yield (**29**). Alkyl substituents on the alkyne were also tolerated yielding all-aliphatic allenes in fair yields (**30**−**31** and **33**−**34**). Secondary propargyl carbonates also participated in the coupling reaction, affording trisubstituted allenes in satisfying yields (**32**−**33**). Notably, the dtbbpy ligand ($L_2$) was more effective than $L_1$ in promoting these reactions. Unfortunately, the more challenging primary carbonate (**34**) afforded only a low yield of the desired terminal allene (23% yield, 70% conversion). Similarly, the reaction with a terminal propargylic electrophile (**35**) resulted in a low yield of the coupling product (10% yield, full conversion). In both cases, the consumption of starting materials was largely accompanied by degradation rather than productive coupling.

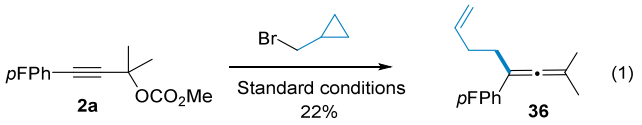
Finally, to elucidate the reaction mechanism, we conducted several mechanistic experiments (see the Supporting Information for details). The reaction was anticipated to proceed through oxidative addition of the propargyl carbonate to nickel, capture of the alkyl radical, and subsequent reductive elimination. We first investigated the nature and oxidation state of the low-valent nickel species purportedly promoting the oxidative addition of the propargyl carbonate. For that purpose, we first evaluated the performance of zerovalent nickel as a catalyst in the synthesis of allene **16**. For instance, under standard conditions, using 1.5 mol % of $Ni(COD)_2/L_1$ as catalyst, the reaction afforded allene **16** in only 25% yield, demonstrating the lower efficiency of zerovalent nickel catalyst in promoting the coupling process.[16]

Next, cyclic voltammetry (CV) studies were conducted to gain further insights into the nickel species responsible for the activation of the propargyl carbonate (**2a**). The cyclic voltammogram of $NiCl_2$·dme/4,4′-$CF_3$-bpy ($L_1$) mixture in 1:1 ratio exhibited two quasi-reversible reduction waves at −1.44 V and −1.78 V (vs $Ag/AgNO_3$) associated with Ni(II)/Ni(I) and Ni(I)/Ni(0) reductions, respectively. The addition of several equivalents of propargyl carbonate (**2a**) (5 and 10 equiv) resulted in an increased current at the first reduction peak Ni(II)/Ni(I) along with a complete loss of



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reversibility. This finding suggests a facile oxidative addition of the propargyl carbonate to photogenerated Ni(I) species. To confirm the involvement of a radical process, 2,2,6,6-tetramethylpiperidinyloxy (TEMPO) was added into the model reaction, which completely suppressed product formation. However, we were unable to observe any TEMPO-adduct. In addition, a radical clock experiment was performed using cyclopropyl methyl bromide as a coupling partner, which provided the alkenyl allene derivative (**36**) as the sole coupling product, indicating the ring opening of a cyclopropyl methyl radical to but-3-enyl radical (eq 1).




Based on documented studies and our observations, the following sequence of events may be proposed as a possible reaction mechanism. The $(TMS)_3SiH$ ($E^{ox}$ [TTMSS/TTMSS$^{\bullet+}$ (**A**)] = +0.73 V vs SCE)[17] would undergo direct oxidative scission of the Si−H bond by the strongly oxidizing photoexcited Ir(III)* photocatalyst ($E_{1/2}^{red}$ [Ir(III)*/Ir(II)] = +1.21 V vs SCE)[18] in the presence of a base to generate a $(TMS)_3Si$ radical intermediate (**B**). This radical would then abstract the bromine atom from the organic halide, yielding the corresponding nucleophilic alkyl radical.[19,20] Concurrently, a low-valent Ni(I) species (**C**), formed via the photocatalytic single-electron reduction of the bipyridyl $NiCl_2$ precatalyst, would undergo $S_N2'$ oxidative addition of the propargyl carbonate to generate allenyl-Ni(III) species (**D**). Subsequently, single electron reduction of this high-valent Ni species is proposed to occur via comproportionation with the remaining Ni(I), yielding the corresponding allenylNi(II) complex (**E**).[21] Capture of the alkyl radical by the Ni(II) species, followed by reductive elimination of the desired alkyl allene from Ni(III) species (**F**) would recycle Ni(I) (Scheme 2). The involvement of Ni(0) species instead of Ni(I) species remains a possibility and could not be completely ruled out. Evidence supporting the involvement of $(TMS)_3SiH$ in bromine abstraction from the alkyl halide was obtained through the detection of methoxytris(trimethylsilyl)silane as a coproduct of the reaction (via GC-MS). This likely arises from nucleophilic substitution of the silyl bromide by methoxide anion released in the reaction medium.[22]

In conclusion, we have developed a new dual nickel/photoredox catalyzed cross-electrophile coupling reaction for the allenylation of C($sp^3$)-X bonds using readily available propargyl carbonates. This methodology operates under very mild and operationally simple conditions, on the benchtop, with low loadings of nickel and photocatalyst under visible light irradiation, and in a nontoxic, sustainable solvent, acetone. The reaction showcases a broad substrate scope, encompassing diverse propargyl carbonates and various primary and secondary alkyl bromides, including elaborated alkyl bromides.

## ■ ASSOCIATED CONTENT

### Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c03512.

Experimental procedures, compound characterization data, and NMR spectra (PDF)

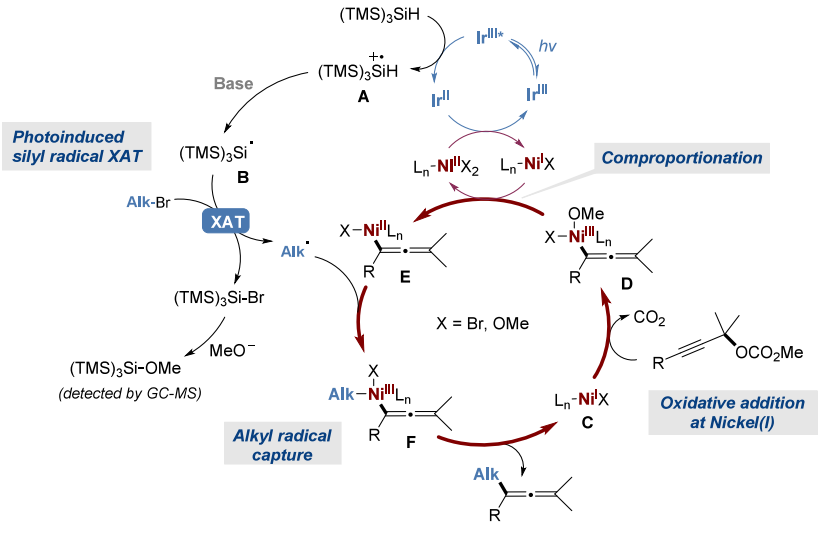
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**Scheme 2. Mechanism Proposal**






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## Author Contributions

[§]T.H. and C.B. contributed equally.

## Notes

The authors declare no competing financial interest.

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## ■ REFERENCES

(1) *Modern Allene Chemistry*; Krause, N., Hashmi, A. S. K., Eds; Wiley-VCH: Weinheim, 2004.

(2) Ma, S. Some typical advances in the synthetic applications of allenes. *Chem. Rev.* **2005**, *105*, 2829−2871.

(3) Hoffmann-Röder, A.; Krause, N. Synthesis and properties of allenic natural products and pharmaceuticals. *Angew. Chem., Int. Ed.* **2004**, *43*, 1196−1216.

(4) Rivera-Fuentes, P.; Diederich, F. Allenes in molecular materials. *Angew. Chem., Int. Ed.* **2012**, *51*, 2818−2828.

(5) Yu, S.; Ma, S. How easy are the syntheses of allenes? *Chem. Commun.* **2011**, *47*, 5384−5418.

(6) (a) Tsuji, J.; Mandai, T. Palladium-catalyzed reactions of propargylic compounds in organic synthesis. *Angew. Chem., Int. Ed. Engl.* **1996**, *34*, 2589−2612. (b) Ma, S. Pd-catalyzed coupling reactions involving propargylic/allenylic species. *Eur. J. Org. Chem.* **2004**, *2004*, 1175−1183. (c) O'Broin, C. Q.; Guiry, P. J. Advances in decarboxylative palladium-catalyzed reactions of propargyl electrophiles. *J. Org. Chem.* **2020**, *85*, 10321−10333.

(7) Selected papers: (a) Jeffery-Luong, T.; Linstrumelle, G. Palladium-catalyzed synthesis of allenes. *Tetrahedron Lett.* **1980**, *21*, 5019−5020. (b) Fürstner, A.; Méndez, M. Iron-catalyzed cross-coupling reactions: Efficient synthesis of 2,3-allenol derivatives. *Angew. Chem., Int. Ed.* **2003**, *42*, 5355−5357. (c) Guisán-Ceinos, M.; Martín-Heras, V.; Soler-Yanes, R.; Cardenas, D. J.; Tortosa, M. Copper-catalysed cross-coupling of alkyl Grignard reagents and propargylic ammonium salts: Stereospecific synthesis of allenes. *Chem. Commun.* **2018**, *54*, 8343−8346. (d) Posevins, D.; Bermejo-López, A.; Bäckvall, J.-E. Iron-catalyzed cross-coupling of propargyl ethers with Grignard reagents for the synthesis of functionalized allenes and allenols. *Angew. Chem., Int. Ed.* **2021**, *60*, 22178−22183.

(8) Selected papers: (a) Soler-Yanes, R.; Arribas-Álvarez, I.; Guisán-Ceinos, M.; Buñuel, E.; Cárdenas, D. J. $Ni^{I}$ catalyzes the regioselective cross-coupling of alkylzinc halides and propargyl bromides to allenes. *Chem.−Eur. J.* **2017**, *23*, 1584−1590. (b) Xu, Y.; Yi, H.; Oestreich, M. Enantioconvergent and regioselective synthesis of allenylsilanes by nickel-catalyzed $C(sp^2)$-$C(sp^3)$ cross-coupling starting from racemic $\alpha$-silylated propargylic bromides. *Organometallics* **2021**, *40*, 2194−2197. (c) Liu, X.; Jiao, C.; Cui, S.; Liu, Q.; Zhang, X.; Zhang, G. Recent advances in nickel-catalyzed synthesis of allenes: New insights and perspectives. *ChemCatChem.* **2023**, *15*, e202300601. (review)

(9) Jin, Y.; Wen, H.; Yang, F.; Ding, D.; Wang, C. Synthesis of multisubstituted allenes via nickel-catalyzed cross-electrophile coupling. *ACS Catal.* **2021**, *11*, 13355−13362.

(10) Review Juliá, F.; Constantin, T.; Leonori, D. Applications of halogen-atom transfer (XAT) for the generation of carbon radicals in synthetic photochemistry and photocatalysis. *Chem. Rev.* **2022**, *122*, 2292−2352.

(11) It is noteworthy that previous metallaphotoredox-catalyzed allenylations of $C(sp^3)$ centres have shown low efficiency for the production of tetrasubstituted allenes: (a) Zhou, Z.-Z.; Song, X.-R.; Du, S.; Xia, K.-J.; Tian, W.-F.; Xiao, Q.; Liang, Y.-M. Photoredox/nickel dual-catalyzed regioselective alkylation of propargylic carbonates for trisubstituted allenes. *Chem. Commun.* **2021**, *57*, 9390−9393. (b) Zhou, Z.-Z.; Zhai, X.-F.; Zhang, S.-L.; Xia, K.-J.; Ding, H.; Song, X.-R.; Tian, W.-F.; Liang, Y.-M.; Xiao, Q. Photo-induced nickel-mediated cross-electrophile coupling for alkylated allenes via electron donor−acceptor complexes. *Org. Chem. Front.* **2023**, *10*, 298−303. (c) Zhou, Z.-Z.; Zhai, X.-F.; Xia, K.-J.; Song, X.-R.; Ding, H.; Tian, W.-F.; Zhang, S.-L.; Liang, Y.-M.; Xiao, Q. Photoredox/nickel dual-catalyzed deaminative cross-electrophile for allenylic alkylation with non-activated alkyl Katritzky salts. *Org. Chem. Front.* **2024**, *11*, 5685.

(12) Chatgilialoglu, C.; Ferreri, C.; Landais, Y.; Timokhin, V. I. Thirty years of $(TMS)_3SiH$: A milestone in radical-based synthetic chemistry. *Chem. Rev.* **2018**, *118*, 6516−6572.

(13) Using the 40 W blue LED lamp at only 50% light intensity proved effective in restraining photodegradation of the allenic products caused by long-time irradiation; see the SI.

(14) The Ni-catalyzed photoredox reduction of propargyl electrophiles with amines is a known process. Hu, T.; Fagué, V.; Bouyssi, D.; Monteiro, N.; Amgoune, A. Hydride-free reduction of propargyl electrophiles: a nickel-catalyzed photoredox strategy for allene synthesis. *Green Chem.* **2024**, *26*, 6124−6130.

(15) Note that $TMS_3SiNH$-(adamantan-1-yl), an XAT agent capable of promoting the activation of alkyl chlorides, was not effective. Sakai, H. A.; Liu, W.; Le, C.; MacMillan, D. W. C. Cross-electrophile coupling of unactivated alkyl chlorides. *J. Am. Chem. Soc.* **2020**, *142*, 11691−11697.

(16) Note that treatment of **2a** with 1 equiv of $L_1/Ni(COD)_2$ leads to full consumption of the propargyl carbonate through homocoupling reaction, which supports facile oxidative addition to Ni(0) species.

(17) Zhu, J.; Cui, W.-C.; Wang, S.; Yao, Z.-J. Radical hydrosilylation of alkynes catalyzed by Eosin Y and thiol under visible light irradiation. *Org. Lett.* **2018**, *20*, 3174−3178.

(18) Lowry, M. S.; Goldsmith, J. I.; Slinker, J. D.; Rohl, R.; Pascal, R. A., Jr.; Malliaras, G. G.; Bernhard, S. Single-layer electroluminescent devices and photoinduced hydrogen production from an ionic Iridium(III) complex. *Chem. Mater.* **2005**, *17*, 5712−5719.

(19) (a) Zhao, X.; MacMillan, D. W. C. Metallaphotoredox perfluoroalkylation of organobromides. *J. Am. Chem. Soc.* **2020**, *142*, 19480−19486. (b) Hell, S. M.; Meyer, C. F.; Laudadio, G.; Misale,

A.; Willis, M. C.; Noël, T.; Trabanco, A. A.; Gouverneur, V. Silyl radical-mediated activation of sulfamoyl chlorides enables direct access to aliphatic sulfonamides from alkenes. *J. Am. Chem. Soc.* **2020**, *142*, 720−725. (c) Kerackian, T.; Reina, A.; Bouyssi, D.; Monteiro, N.; Amgoune, A. Silyl radical mediated cross-electrophile coupling of *N*-acyl-imides with alkyl bromides under photoredox/nicked dual catalysis. *Org. Lett.* **2020**, *22*, 2240−2245.

(20) At this stage we cannot dismiss possible generation of a bromine radical, via single electron oxidation of released $Br^-$ anion by the photoexcited *Ir(III) catalyst, as an alternative pathway to silyl radical formation via hydrogen atom abstraction from $(TMS)_3SiH$. Zhang, P.; Le, C.; MacMillan, D. W. C. Silyl radical activation of alkyl halides in metallaphotoredox catalysis: A unique pathway for cross-electrophile coupling. *J. Am. Chem. Soc.* **2016**, *138*, 8084−8087.

(21) (a) Day, C. S.; Martin, R. Comproportionation and disproportionation in nickel and copper complexes. *Chem. Soc. Rev.* **2023**, *52*, 6601−6616. (b) Cusumano, A. Q.; Chaffin, B. C.; Doyle, A. G. Mechanism of Ni-catalyzed photochemical halogen atom-mediated C($sp^3$)-H arylation. *J. Am. Chem. Soc.* **2024**, *146*, 15331−15344.

(22) Marsmann, H. C.; Raml, W.; Hengge, E. $^{29}Si$ NMR Measurements on Polysilanes. 2. Isotetrasilanes. *Z. Naturforsch. B* **1980**, *35*, 1541−1547.



https://doi.org/10.1021/acs.orglett.4c03512
*Org. Lett.* 2024, 26, 9519−9524